

## 122. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8

### 122.1 Overview

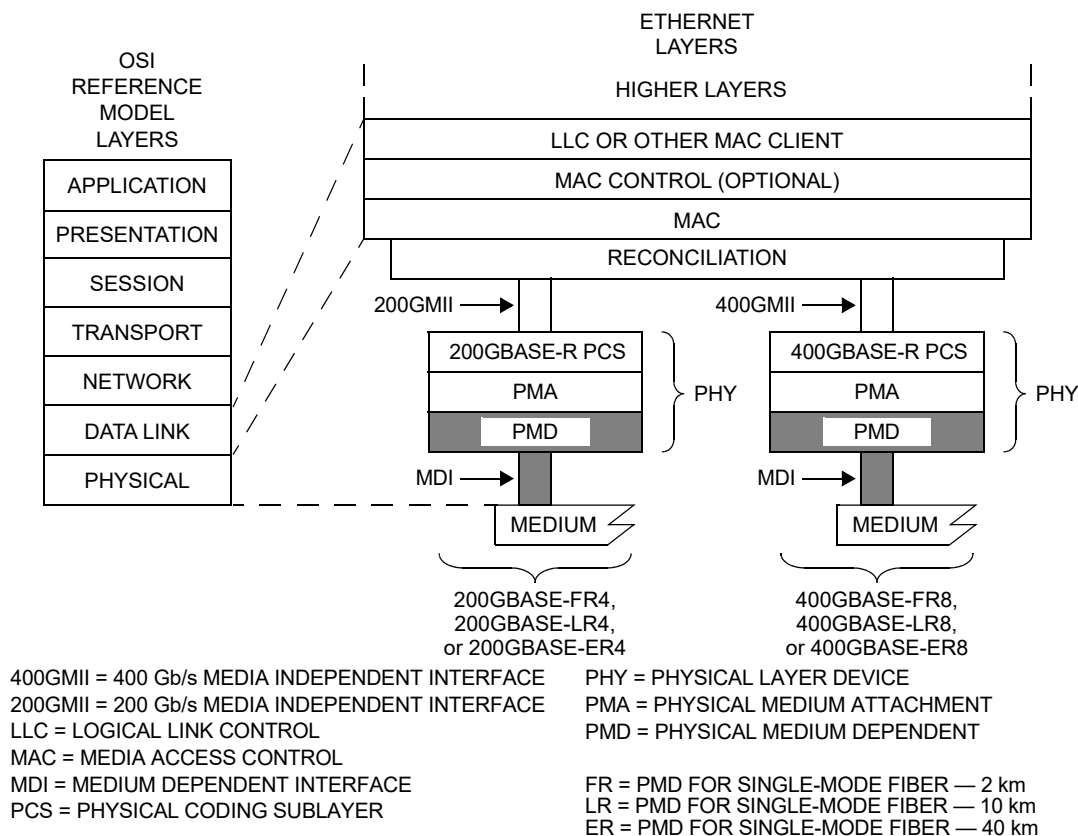
This clause specifies the 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs together with the single-mode fiber medium. The optical signals generated by these six PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 122–1, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

**Table 122–1—Physical Layer clauses associated with the 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs**

| Associated clause              | 200GBASE-FR4,<br>200GBASE-LR4,<br>200GBASE-ER4 | 400GBASE-FR8,<br>400GBASE-LR8,<br>400GBASE-ER8 |
|--------------------------------|------------------------------------------------|------------------------------------------------|
| 117—RS                         | Required                                       | Required                                       |
| 117—200GMII <sup>a</sup>       | Optional                                       | Not applicable                                 |
| 117—400GMII <sup>a</sup>       | Not applicable                                 | Optional                                       |
| 118—200GMII Extender           | Optional                                       | Not applicable                                 |
| 118—400GMII Extender           | Not applicable                                 | Optional                                       |
| 119—PCS for 200GBASE-R         | Required                                       | Not applicable                                 |
| 119—PCS for 400GBASE-R         | Not applicable                                 | Required                                       |
| 120—PMA for 200GBASE-R         | Required                                       | Not applicable                                 |
| 120—PMA for 400GBASE-R         | Not applicable                                 | Required                                       |
| 120B—Chip-to-chip 200GAUI-8    | Optional                                       | Not applicable                                 |
| 120B—Chip-to-chip 400GAUI-16   | Not applicable                                 | Optional                                       |
| 120C—Chip-to-module 200GAUI-8  | Optional                                       | Not applicable                                 |
| 120C—Chip-to-module 400GAUI-16 | Not applicable                                 | Optional                                       |
| 120D—Chip-to-chip 200GAUI-4    | Optional                                       | Not applicable                                 |
| 120D—Chip-to-chip 400GAUI-8    | Not applicable                                 | Optional                                       |
| 120E—Chip-to-module 200GAUI-4  | Optional                                       | Not applicable                                 |
| 120E—Chip-to-module 400GAUI-8  | Not applicable                                 | Optional                                       |
| 78—Energy Efficient Ethernet   | Optional                                       | Optional                                       |

<sup>a</sup> 200GMII and 400GMII are optional interfaces. However, if the appropriate interface is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII or 400GMII were present.

Figure 122–1 shows the relationship of the PMD and MDI (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 200 Gb/s and 400 Gb/s Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2.



**Figure 122–1—200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model**

200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 use four lanes, while 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 use eight lanes. In this clause, where there are four or eight items (depending on PMD type) such as lanes, the items are numbered from 0 to  $n - 1$ , and an example item is numbered  $i$ . Thus  $n$  is 4 or 8.

200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PHYs with the optional Energy Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

### 122.1.1 Bit error ratio

The bit error ratio (BER) when processed according to Clause 120 shall be less than  $2.4 \times 10^{-4}$  provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than  $1.7 \times 10^{-12}$  for 64-octet frames with minimum interpacket gap when processed according to Clause 120 and then Clause 119. For a complete Physical Layer, the frame loss ratio may be degraded to  $6.2 \times 10^{-11}$  for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces.

If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than  $1.7 \times 10^{-12}$  for 64-octet frames with minimum interpacket gap.

## 122.2 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3. The PMD service interface primitives are summarized as follows:

PMD:IS\_UNITDATA\_*i*.request  
PMD:IS\_UNITDATA\_*i*.indication  
PMD:IS\_SIGNAL.indication

The 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 PMDs have four parallel symbol streams, in which case  $i = 0$  to 3, and the 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs have eight parallel symbol streams, in which case  $i = 0$  to 7.

In the transmit direction, the PMA continuously sends  $n$  parallel symbol streams to the PMD, one per lane, each at a nominal signaling rate of 26.5625 GBd. The PMD then converts these streams of data units into the appropriate signals on the MDI.

In the receive direction, the PMD continuously sends  $n$  parallel symbol streams to the PMA corresponding to the signals received from the MDI, one per lane, each at a nominal signaling rate of 26.5625 GBd.

The SIGNAL\_DETECT parameter defined in this clause maps to the SIGNAL\_OK parameter in the PMD:IS\_SIGNAL.indication(SIGNAL\_OK) inter-sublayer service interface primitive defined in 116.3.

The SIGNAL\_DETECT parameter can take on one of two values: OK or FAIL. When SIGNAL\_DETECT = FAIL, the rx\_symbol parameters are undefined.

NOTE—SIGNAL\_DETECT = OK does not guarantee that the rx\_symbol parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL\_DETECT = OK indication and still not meet the BER defined in 122.1.1.

## 122.3 Delay and Skew

### 122.3.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-FR4, 200GBASE-LR4, or 200GBASE-ER4 PMD including 2 m of fiber in one direction shall be no more than 4096 bit times (8 pause\_quanta or 20.48 ns). The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause\_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause\_quanta can be found in 116.4 and its references.

### 122.3.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS. Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116-4 and Figure 116-5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5. The measurements of Skew and Skew Variation are defined in 87.8.2.

### 122.4 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that may provide control and status information for and about the PMD. If the MDIO interface is implemented, the mapping of MDIO control variables to PMD control variables shall be as shown in Table 122-2 and the mapping of MDIO status variables to PMD status variables shall be as shown in Table 122-3.

**Table 122-2—MDIO/PMD control variable mapping**

| MDIO control variable                                | PMA/PMD register name         | Register/bit number | PMD control variable                                 |
|------------------------------------------------------|-------------------------------|---------------------|------------------------------------------------------|
| Reset                                                | PMA/PMD control 1 register    | 1.0.15              | PMD_reset                                            |
| Global PMD transmit disable                          | PMD transmit disable register | 1.9.0               | PMD_global_transmit_disable                          |
| PMD transmit disable $n-1$ to PMD transmit disable 0 | PMD transmit disable register | 1.9.8 to 1.9.1      | PMD_transmit_disable $n-1$ to PMD_transmit_disable_0 |

### 122.5 PMD functional specifications

The 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

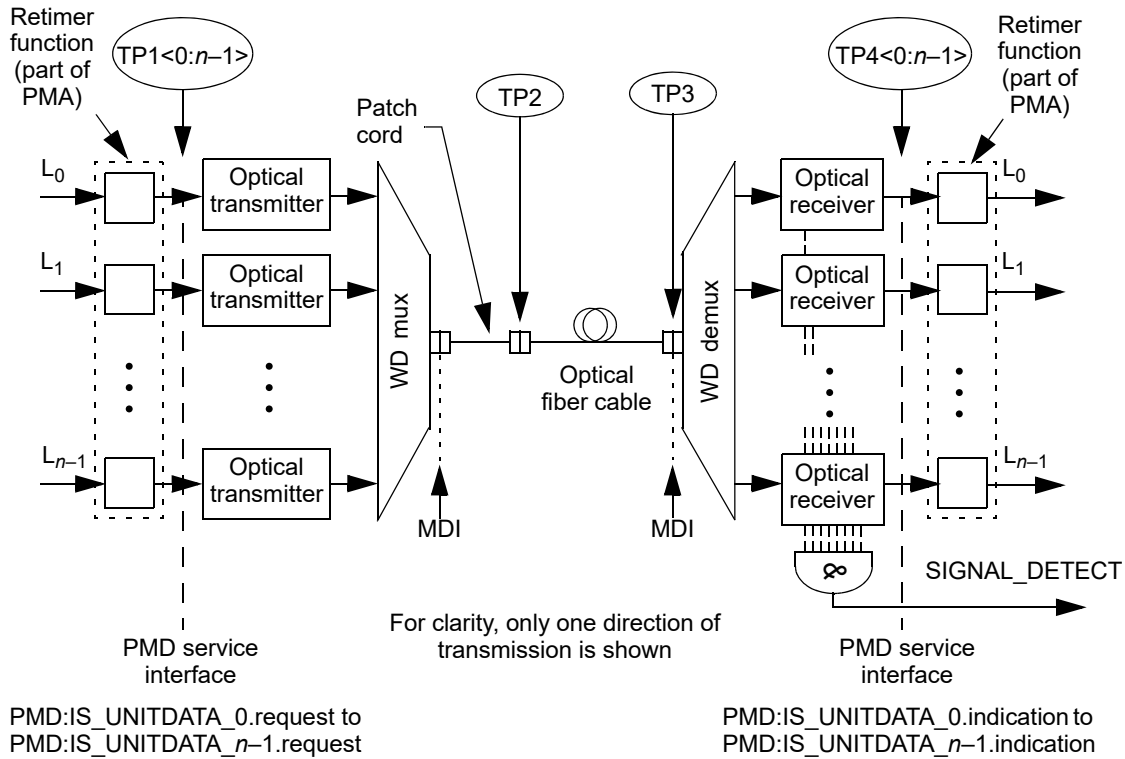
#### 122.5.1 PMD block diagram

The PMD block diagram is shown in Figure 122-2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 122.8 are made at TP2. The optical receive signal is defined at

**Table 122–3—MDIO/PMD status variable mapping**

| MDIO status variable                                              | PMA/PMD register name              | Register/<br>bit number | PMD status variable                                |
|-------------------------------------------------------------------|------------------------------------|-------------------------|----------------------------------------------------|
| Fault                                                             | PMA/PMD status 1 register          | 1.1.7                   | PMD_fault                                          |
| Transmit fault                                                    | PMA/PMD status 2 register          | 1.8.11                  | PMD_transmit_fault                                 |
| Receive fault                                                     | PMA/PMD status 2 register          | 1.8.10                  | PMD_receive_fault                                  |
| Global PMD receive signal detect                                  | PMD receive signal detect register | 1.10.0                  | PMD_global_signal_detect                           |
| PMD receive signal detect $n-1$ to<br>PMD receive signal detect 0 | PMD receive signal detect register | 1.10.8 to<br>1.10.1     | PMD_signal_detect_ $n-1$ to<br>PMD_signal_detect_0 |

the output of the fiber optic cabling (TP3) at the MDI (see 122.11.3). Unless specified otherwise, all receiver measurements and tests defined in 122.8 are made at TP3.



WD = Wavelength division

NOTE—Specification of the retimer function and the electrical implementation of the PMD service interface is beyond the scope of this standard.

**Figure 122–2—Block diagram for 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 transmit/receive paths**

TP1<0:n-1> and TP4<0:n-1> are optional reference points that may be useful to implementers for testing components (these test points will not typically be accessible in an implemented system).

### 122.5.2 PMD transmit function

The PMD Transmit function shall convert the  $n$  symbol streams requested by the PMD service interface messages PMD:IS\_UNITDATA\_0.request to PMD:IS\_UNITDATA\_ $n-1$ .request into  $n$  separate optical signals. The  $n$  optical signals shall then be wavelength division multiplexed and delivered to the MDI, all according to the transmit optical specifications in this clause. The highest optical power level in each signal shall correspond to tx\_symbol = three and the lowest shall correspond to tx\_symbol = zero.

### 122.5.3 PMD receive function

The PMD Receive function shall demultiplex the composite optical signal received from the MDI into  $n$  separate optical signals. The  $n$  optical signals shall then be converted into  $n$  symbol streams for delivery to the PMD service interface using the messages PMD:IS\_UNITDATA\_0.indication to PMD:IS\_UNITDATA\_ $n-1$ .indication, all according to the receive optical specifications in this clause. The higher optical power level in each signal shall correspond to rx\_symbol = three and the lowest shall correspond to rx\_symbol = zero.

### 122.5.4 PMD global signal detect function

The PMD global signal detect function shall report the state of SIGNAL\_DETECT via the PMD service interface. The SIGNAL\_DETECT parameter is signaled continuously, while the PMD:IS\_SIGNAL.indication message is generated when a change in the value of SIGNAL\_DETECT occurs. The SIGNAL\_DETECT parameter defined in this clause maps to the SIGNAL\_OK parameter in the inter-sublayer service interface primitives defined in 116.3.

SIGNAL\_DETECT shall be a global indicator of the presence of optical signals on all  $n$  lanes. The value of the SIGNAL\_DETECT parameter shall be generated according to the conditions defined in Table 122–4. The PMD receiver is not required to verify whether a compliant 200GBASE-R or 400GBASE-R signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL\_DETECT parameter.

**Table 122–4—SIGNAL\_DETECT value definition**

| Receive conditions                                                                                                                                                                                                                                                                     | SIGNAL_DETECT value |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| For any lane; Average optical power at TP3 $\leq -30$ dBm                                                                                                                                                                                                                              | FAIL                |
| For all lanes;<br>[(Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 122–11 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 or Table 122–12 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8)<br>AND<br>(compliant 200GBASE-R or 400GBASE-R signal input)] | OK                  |
| All other conditions                                                                                                                                                                                                                                                                   | Unspecified         |

As a consequence of the requirements for the setting of the SIGNAL\_DETECT parameter, implementations need to provide adequate margin between the input optical power level at which the SIGNAL\_DETECT parameter is set to OK, and the inherent noise level of the PMD including the effects of crosstalk, power supply noise, etc.

Various implementations of the Signal Detect function are permitted by this standard, including implementations that generate the SIGNAL\_DETECT parameter values in response to the amplitude of the

modulation of the optical signal and implementations that respond to the average optical power of the modulated optical signal.

#### 122.5.5 PMD lane-by-lane signal detect function

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each `PMD_signal_detect_i`, where  $i$  represents the lane number in the range  $0:n-1$ , shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 122-4.

#### 122.5.6 PMD reset function

If the MDIO interface is implemented, and if `PMD_reset` is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

#### 122.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

- a) When the `PMD_global_transmit_disable` variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 122-9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and Table 122-10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.
- b) If a `PMD_fault` is detected, then the PMD may set the `PMD_global_transmit_disable` variable to one, turning off the optical transmitter in each lane.

#### 122.5.8 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function allows the optical transmitters in each lane to be selectively disabled.

- a) When a `PMD_transmit_disable_i` variable (where  $i$  represents the lane number in the range  $0:3$  for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and  $0:7$  for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 122-9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and Table 122-10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.
- b) If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disable_i` variable to one, turning off the optical transmitter in each lane.

If the PMD lane-by-lane transmit disable function is not implemented in MDIO, an alternative method shall be provided to independently disable each transmit lane for testing purposes.

#### 122.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set `PMD_fault` to one.

If the MDIO interface is implemented, `PMD_fault` shall be mapped to the fault bit as specified in 45.2.1.2.3.

#### 122.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the `PMD_transmit_fault` variable to one.

If the MDIO interface is implemented, PMD\_transmit\_fault shall be mapped to the transmit fault bit as specified in 45.2.1.7.4.

#### 122.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD\_receive\_fault variable to one.

If the MDIO interface is implemented, PMD\_receive\_fault shall be mapped to the receive fault bit as specified in 45.2.1.7.5.

### 122.6 Wavelength-division-multiplexed lane assignments

The wavelength range for each lane of the 200GBASE-FR4 PMD is defined in Table 122–5. The 200GBASE-FR4 center wavelengths are members of the CWDM wavelength grid defined in ITU-T G.694.2 and are spaced at 20 nm.

The wavelength range for each lane of the 200GBASE-LR4 and 200GBASE-ER4 PMDs is defined in Table 122–6. The wavelength range for each lane of the 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs is defined in Table 122–7. The 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 center frequencies are members of the frequency grid for 100 GHz spacing and above defined in ITU-T G.694.1 and are spaced at multiples of 800 GHz.

**Table 122–5—200GBASE-FR4 wavelength-division-multiplexed lane assignments**

| Lane           | Center wavelength | Wavelength range    |
|----------------|-------------------|---------------------|
| L <sub>0</sub> | 1271 nm           | 1264.5 to 1277.5 nm |
| L <sub>1</sub> | 1291 nm           | 1284.5 to 1297.5 nm |
| L <sub>2</sub> | 1311 nm           | 1304.5 to 1317.5 nm |
| L <sub>3</sub> | 1331 nm           | 1324.5 to 1337.5 nm |

**Table 122–6—200GBASE-LR4 and 200GBASE-ER4 wavelength-division-multiplexed lane assignments**

| Lane           | Center frequency | Center wavelength | Wavelength range      |
|----------------|------------------|-------------------|-----------------------|
| L <sub>0</sub> | 231.4 THz        | 1295.56 nm        | 1294.53 to 1296.59 nm |
| L <sub>1</sub> | 230.6 THz        | 1300.05 nm        | 1299.02 to 1301.09 nm |
| L <sub>2</sub> | 229.8 THz        | 1304.58 nm        | 1303.54 to 1305.63 nm |
| L <sub>3</sub> | 229 THz          | 1309.14 nm        | 1308.09 to 1310.19 nm |

NOTE—There is no requirement to associate a particular electrical lane with a particular optical lane, as the PCS is capable of receiving lanes in any arrangement.

**Table 122–7—400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8  
wavelength-division-multiplexed lane assignments**

| Lane           | Center frequency | Center wavelength | Wavelength range      |
|----------------|------------------|-------------------|-----------------------|
| L <sub>0</sub> | 235.4 THz        | 1273.54 nm        | 1272.55 to 1274.54 nm |
| L <sub>1</sub> | 234.6 THz        | 1277.89 nm        | 1276.89 to 1278.89 nm |
| L <sub>2</sub> | 233.8 THz        | 1282.26 nm        | 1281.25 to 1283.27 nm |
| L <sub>3</sub> | 233 THz          | 1286.66 nm        | 1285.65 to 1287.68 nm |
| L <sub>4</sub> | 231.4 THz        | 1295.56 nm        | 1294.53 to 1296.59 nm |
| L <sub>5</sub> | 230.6 THz        | 1300.05 nm        | 1299.02 to 1301.09 nm |
| L <sub>6</sub> | 229.8 THz        | 1304.58 nm        | 1303.54 to 1305.63 nm |
| L <sub>7</sub> | 229 THz          | 1309.14 nm        | 1308.09 to 1310.19 nm |

### **122.7 PMD to MDI optical specifications for 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8**

The operating ranges for the 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs are defined in Table 122–8. A 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 compliant PMD operates on type B1.1, B1.3, or B6 single-mode fibers according to the specifications defined in Table 122–18. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-FR8 PMD operating at 2.5 km meets the operating range requirement of 2 m to 2 km).

The 200GBASE-ER4 PMD interoperates with the 200GBASE-LR4 PMD provided that the channel requirements defined in 122.12 are met.

The 400GBASE-LR8 PMD interoperates with the 400GBASE-FR8 PMD provided that the channel requirements for 400GBASE-FR8 are met.

The 400GBASE-ER8 PMD interoperates with the 400GBASE-FR8 PMD provided that the channel requirements defined in 122.13 are met.

The 400GBASE-ER8 PMD interoperates with the 400GBASE-LR8 PMD provided that the channel requirements defined in 122.14 are met.

**Table 122–8—200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 operating ranges**

| PMD type                      | Required operating range  |
|-------------------------------|---------------------------|
| 200GBASE-FR4 and 400GBASE-FR8 | 2 m to 2 km               |
| 200GBASE-LR4 and 400GBASE-LR8 | 2 m to 10 km              |
| 200GBASE-ER4 and 400GBASE-ER8 | 2 m to 30 km              |
|                               | 2 m to 40 km <sup>a</sup> |

<sup>a</sup> Links longer than 30 km for the same link power budget are considered engineered links. Attenuation for such links needs to be less than the worst case specified for IEC 60793-2-50 type B1.1, type B1.3, or type B6\_a single-mode fiber.

#### **122.7.1 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 transmitter optical specifications**

The 200GBASE-FR4 transmitter shall meet the specifications defined in Table 122–9 per the definitions in 122.8. The 200GBASE-LR4 transmitter shall meet the specifications defined in Table 122–9 per the definitions in 122.8. The 200GBASE-ER4 transmitter shall meet the specifications defined in Table 122–9 per the definitions in 122.8. The 400GBASE-FR8 transmitter shall meet the specifications defined in Table 122–10 per the definitions in 122.8. The 400GBASE-LR8 transmitter shall meet the specifications defined in Table 122–10 per the definitions in 122.8. The 400GBASE-ER8 transmitter shall meet the specifications defined in Table 122–10 per the definitions in 122.8.

**Table 122–9—200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 transmit characteristics**

| Description                                                                                                                          | 200GBASE-FR4                                                                 | 200GBASE-LR4                                                                         | 200GBASE-ER4 | Unit       |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------|------------|
| Signaling rate, each lane (range)                                                                                                    | 26.5625 ± 100 ppm                                                            |                                                                                      |              | GBd        |
| Modulation format                                                                                                                    | PAM4                                                                         |                                                                                      |              | —          |
| Lane wavelengths (range)                                                                                                             | 1264.5 to 1277.5<br>1284.5 to 1297.5<br>1304.5 to 1317.5<br>1324.5 to 1337.5 | 1294.53 to 1296.59<br>1299.02 to 1301.09<br>1303.54 to 1305.63<br>1308.09 to 1310.19 |              | nm         |
| Side-mode suppression ratio (SMSR), (min)                                                                                            | 30                                                                           |                                                                                      |              | dB         |
| Total average launch power (max)                                                                                                     | 10.7                                                                         | 11.3                                                                                 | 12.6         | dBm        |
| Average launch power, each lane (max)                                                                                                | 4.7                                                                          | 5.3                                                                                  | 6.6          | dBm        |
| Average launch power, each lane <sup>a</sup> (min)                                                                                   | −4.2                                                                         | −3.4                                                                                 | 0.4          | dBm        |
| Outer Optical Modulation Amplitude (OMA <sub>outer</sub> ), each lane (max)                                                          | 4.5                                                                          | 5.1                                                                                  | 7.4          | dBm        |
| Outer Optical Modulation Amplitude (OMA <sub>outer</sub> ), each lane (min) <sup>b</sup>                                             | −1.2                                                                         | −0.4                                                                                 | 3.4          | dBm        |
| Difference in launch power between any two lanes (OMA <sub>outer</sub> ) (max)                                                       | 4                                                                            |                                                                                      |              | dB         |
| Launch power in OMA <sub>outer</sub> minus TDECQ, each lane (min):<br>for extinction ratio ≥ 4.5 dB<br>for extinction ratio < 4.5 dB | −2.6<br>−2.5                                                                 | −1.8<br>−1.7                                                                         | 2<br>—       | dBm<br>dBm |
| Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)                                                             | 3.1                                                                          | 3.2                                                                                  | 3.2          | dB         |
| TDECQ −10log <sub>10</sub> (C <sub>eq</sub> ) <sup>c</sup> (max)                                                                     | 3.1                                                                          | 3.2                                                                                  | 3.2          | dB         |
| Average launch power of OFF transmitter, each lane (max)                                                                             | −30                                                                          |                                                                                      |              | dBm        |
| Extinction ratio, each lane (min)                                                                                                    | 3.5                                                                          |                                                                                      | 6            | dB         |
| Transmitter transition time (max)                                                                                                    | 34                                                                           |                                                                                      |              | ps         |
| RIN <sub>17.1</sub> OMA (max)                                                                                                        | −132                                                                         | —                                                                                    | —            | dB/Hz      |
| RIN <sub>15.6</sub> OMA (max)                                                                                                        | —                                                                            | −132                                                                                 | —            | dB/Hz      |
| RIN <sub>15</sub> OMA (max)                                                                                                          | —                                                                            | —                                                                                    | −132         | dB/Hz      |
| Optical return loss tolerance (max)                                                                                                  | 17.1                                                                         | 15.6                                                                                 | 15           | dB         |
| Transmitter reflectance <sup>d</sup> (max)                                                                                           | −26                                                                          |                                                                                      |              | dB         |

<sup>a</sup> Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

<sup>b</sup> Even if the TDECQ < 1.4 dB for an extinction ratio of ≥ 4.5 dB or TDECQ < 1.3 dB for an extinction ratio of < 4.5 dB, the OMA<sub>outer</sub> (min) has to exceed this value.

<sup>c</sup> C<sub>eq</sub> is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

<sup>d</sup> Transmitter reflectance is defined looking into the transmitter.

**Table 122–10—400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 transmit characteristics**

| Description                                                                                                                          | 400GBASE-FR8                                                                                                                                                                 | 400GBASE-LR8 | 400GBASE-ER8 | Unit  |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-------|
| Signaling rate, each lane (range)                                                                                                    | 26.5625 ± 100 ppm                                                                                                                                                            |              |              | GBd   |
| Modulation format                                                                                                                    | PAM4                                                                                                                                                                         |              |              | —     |
| Lane wavelengths (range)                                                                                                             | 1272.55 to 1274.54<br>1276.89 to 1278.89<br>1281.25 to 1283.27<br>1285.65 to 1287.68<br>1294.53 to 1296.59<br>1299.02 to 1301.09<br>1303.54 to 1305.63<br>1308.09 to 1310.19 |              |              | nm    |
| Side-mode suppression ratio (SMSR), (min)                                                                                            | 30                                                                                                                                                                           |              |              | dB    |
| Total average launch power (max)                                                                                                     | 13.2                                                                                                                                                                         |              | 14.6         | dBm   |
| Average launch power, each lane <sup>a</sup> (max)                                                                                   | 5.3                                                                                                                                                                          |              | 5.6          | dBm   |
| Average launch power, each lane <sup>b</sup> (min)                                                                                   | −3.5                                                                                                                                                                         | −2.8         | −0.6         | dBm   |
| Outer Optical Modulation Amplitude (OMA <sub>outer</sub> ), each lane (max)                                                          | 5.5                                                                                                                                                                          | 5.7          | 6.4          | dBm   |
| Outer Optical Modulation Amplitude (OMA <sub>outer</sub> ), each lane (min) <sup>c</sup>                                             | −0.5                                                                                                                                                                         | 0.2          | 2.4          | dBm   |
| Difference in launch power between any two lanes (OMA <sub>outer</sub> ) (max)                                                       | 4                                                                                                                                                                            |              |              | dB    |
| Launch power in OMA <sub>outer</sub> minus TDECQ, each lane (min):<br>for extinction ratio ≥ 4.5 dB<br>for extinction ratio < 4.5 dB | −1.9                                                                                                                                                                         | −1.2         | 1            | dBm   |
|                                                                                                                                      | −1.8                                                                                                                                                                         | −1.1         | —            | dBm   |
| Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)                                                             | 2.9                                                                                                                                                                          | 3.1          | 3.4          | dB    |
| TDECQ −10log <sub>10</sub> (C <sub>eq</sub> ) <sup>d</sup> (max)                                                                     | 2.9                                                                                                                                                                          | 3.1          | 3.4          | dB    |
| Average launch power of OFF transmitter, each lane (max)                                                                             | −30                                                                                                                                                                          |              |              | dBm   |
| Extinction ratio, each lane (min)                                                                                                    | 3.5                                                                                                                                                                          |              | 6            | dB    |
| Transmitter transition time (max)                                                                                                    | 34                                                                                                                                                                           |              |              | ps    |
| RIN <sub>17.1</sub> OMA (max)                                                                                                        | −132                                                                                                                                                                         | —            | —            | dB/Hz |
| RIN <sub>15.6</sub> OMA (max)                                                                                                        | —                                                                                                                                                                            | −132         | —            | dB/Hz |
| RIN <sub>15</sub> OMA (max)                                                                                                          | —                                                                                                                                                                            | —            | −132         | dB/Hz |
| Optical return loss tolerance (max)                                                                                                  | 17.1                                                                                                                                                                         | 15.6         | 15           | dB    |
| Transmitter reflectance <sup>e</sup> (max)                                                                                           | −26                                                                                                                                                                          |              |              | dB    |

<sup>a</sup> As the total average launch power limit has to be met, not all of the lanes can operate at the maximum average launch power, each lane.

<sup>b</sup> Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

<sup>c</sup> Even if the TDECQ < 1.4 dB for an extinction ratio of ≥ 4.5 dB or TDECQ < 1.3 dB for an extinction ratio of < 4.5 dB, the OMA<sub>outer</sub> (min) has to exceed this value.

<sup>d</sup> C<sub>eq</sub> is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

<sup>e</sup> Transmitter reflectance is defined looking into the transmitter.

## 122.7.2 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 receive optical specifications

The 200GBASE-FR4 receiver shall meet the specifications defined in Table 122–11 per the definitions in 122.8. The 200GBASE-LR4 receiver shall meet the specifications defined in Table 122–11 per the definitions in 122.8. The 200GBASE-ER4 receiver shall meet the specifications defined in Table 122–11 per the definitions in 122.8. The 400GBASE-FR8 receiver shall meet the specifications defined in Table 122–12 per the definitions in 122.8. The 400GBASE-LR8 receiver shall meet the specifications defined in Table 122–12 per the definitions in 122.8. The 400GBASE-ER8 receiver shall meet the specifications defined in Table 122–12 per the definitions in 122.8.

**Table 122–11—200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 receive characteristics**

| Description                                                                         | 200GBASE-FR4                                                                 | 200GBASE-LR4                                                                         | 200GBASE-ER4     | Unit |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------|------|
| Signaling rate, each lane (range)                                                   | 26.5625 ± 100 ppm                                                            |                                                                                      |                  | GBd  |
| Modulation format                                                                   | PAM4                                                                         |                                                                                      |                  | —    |
| Lane wavelengths (range)                                                            | 1264.5 to 1277.5<br>1284.5 to 1297.5<br>1304.5 to 1317.5<br>1324.5 to 1337.5 | 1294.53 to 1296.59<br>1299.02 to 1301.09<br>1303.54 to 1305.63<br>1308.09 to 1310.19 |                  | nm   |
| Damage threshold <sup>a</sup> , each lane                                           | 5.7                                                                          | 6.3                                                                                  | –2.4             | dBm  |
| Average receive power, each lane (max)                                              | 4.7                                                                          | 5.3                                                                                  | –3.4             | dBm  |
| Average receive power, each lane <sup>b</sup> (min)                                 | –8.2                                                                         | –9.7                                                                                 | –17.6            | dBm  |
| Receive power (OMA <sub>outer</sub> ), each lane (max)                              | 4.5                                                                          | 5.1                                                                                  | –2.6             | dBm  |
| Difference in receive power between any two lanes (OMA <sub>outer</sub> ) (max)     | 4.1                                                                          | 4.2                                                                                  | 4.6              | dB   |
| Receiver reflectance (max)                                                          | –26                                                                          |                                                                                      |                  | dB   |
| Receiver sensitivity (OMA <sub>outer</sub> ), each lane <sup>c</sup> (max)          | Equation (122–1)                                                             | Equation (122–2)                                                                     | Equation (122–3) | dBm  |
| Stressed receiver sensitivity (OMA <sub>outer</sub> ), each lane <sup>d</sup> (max) | –3.8                                                                         | –5.4                                                                                 | –13.3            | dBm  |
| Conditions of stressed receiver sensitivity test: <sup>e</sup>                      |                                                                              |                                                                                      |                  |      |
| Stressed eye closure for PAM4 (SECQ), lane under test                               | 3.1                                                                          | 3.2                                                                                  | 3.2              | dB   |
| SECQ – 10log <sub>10</sub> (C <sub>eq</sub> ) <sup>f</sup> (max), lane under test   | 3.1                                                                          | 3.2                                                                                  | 3.2              | dB   |
| OMA <sub>outer</sub> of each aggressor lane                                         | 0.3                                                                          | –1.2                                                                                 | –8.7             | dBm  |

<sup>a</sup> The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

<sup>b</sup> Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

<sup>c</sup> Receiver sensitivity (OMA<sub>outer</sub>), each lane (max) is optional and is defined for a transmitter with a value of SECQ up to 3.1 dB for 200GBASE-FR4, 3.2 dB for 200GBASE-LR4, and 3.2 dB for 200GBASE-ER4.

<sup>d</sup> Measured with conformance test signal at TP3 (see 122.8.10) for the BER specified in 122.1.1.

<sup>e</sup> These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

<sup>f</sup> C<sub>eq</sub> is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

**Table 122–12—400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 receive characteristics**

| Description                                                                         | 400GBASE-FR8                                                                                                                                                                 | 400GBASE-LR8     | 400GBASE-ER8     | Unit |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------|------|
| Signaling rate, each lane (range)                                                   | 26.5625 ± 100 ppm                                                                                                                                                            |                  |                  | GBd  |
| Modulation format                                                                   | PAM4                                                                                                                                                                         |                  |                  | —    |
| Lane wavelengths (range)                                                            | 1272.55 to 1274.54<br>1276.89 to 1278.89<br>1281.25 to 1283.27<br>1285.65 to 1287.68<br>1294.53 to 1296.59<br>1299.02 to 1301.09<br>1303.54 to 1305.63<br>1308.09 to 1310.19 |                  |                  | nm   |
| Damage threshold <sup>a</sup> , each lane                                           | 6.3                                                                                                                                                                          |                  | –3.4             | dBm  |
| Average receive power, each lane (max)                                              | 5.3                                                                                                                                                                          |                  | –4.4             | dBm  |
| Average receive power, each lane <sup>b</sup> (min)                                 | –7.5                                                                                                                                                                         | –9.1             | –18.6            | dBm  |
| Receive power (OMA <sub>outer</sub> ), each lane (max)                              | 5.7                                                                                                                                                                          |                  | –3.6             | dBm  |
| Difference in receive power between any two lanes (OMA <sub>outer</sub> ) (max)     | 4.1                                                                                                                                                                          | 4.5              | 5.8              | dB   |
| Receiver reflectance (max)                                                          | –26                                                                                                                                                                          |                  |                  | dB   |
| Receiver sensitivity (OMA <sub>outer</sub> ), each lane <sup>c</sup> (max)          | Equation (122–4)                                                                                                                                                             | Equation (122–5) | Equation (122–6) | dBm  |
| Stressed receiver sensitivity (OMA <sub>outer</sub> ), each lane <sup>d</sup> (max) | –3.3                                                                                                                                                                         | –4.9             | –14.1            | dBm  |
| Conditions of stressed receiver sensitivity test: <sup>e</sup>                      |                                                                                                                                                                              |                  |                  |      |
| Stressed eye closure for PAM4 (SECQ), lane under test                               | 2.9                                                                                                                                                                          | 3.1              | 3.4              | dB   |
| SECQ – 10log <sub>10</sub> (C <sub>eq</sub> ) <sup>f</sup> (max), lane under test   | 2.9                                                                                                                                                                          | 3.1              | 3.4              | dB   |
| OMA <sub>outer</sub> of each aggressor lane                                         | 0.8                                                                                                                                                                          | –0.4             | –8.3             | dBm  |

<sup>a</sup> The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

<sup>b</sup> Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

<sup>c</sup> Receiver sensitivity (OMA<sub>outer</sub>), each lane (max) is optional and is defined for a transmitter with a value of SECQ up to 2.9 dB for 400GBASE-FR8, 3.1 dB for 400GBASE-LR8, and 3.4 dB for 400GBASE-ER8.

<sup>d</sup> Measured with conformance test signal at TP3 (see 122.8.10) for the BER specified in 122.1.1.

<sup>e</sup> These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

<sup>f</sup> C<sub>eq</sub> is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

### 122.7.3 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 illustrative link power budgets

Illustrative power budgets and penalties for 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 channels are shown in Table 122–13.

**Table 122–13—200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 illustrative link power budgets**

| Parameter                                                                                                                    | 200GBASE-FR4   | 400GBASE-FR8 | 200GBASE-LR4 | 400GBASE-LR8 | 200GBASE-ER4 |                 | 400GBASE-ER8 |                 | Unit     |
|------------------------------------------------------------------------------------------------------------------------------|----------------|--------------|--------------|--------------|--------------|-----------------|--------------|-----------------|----------|
| Power budget (for maximum TDECQ):<br>for extinction ratio ≥ 4.5 dB<br>for extinction ratio < 4.5 dB                          | 7.4<br>7.5     | 7.2<br>7.3   | 10<br>10.1   | 9.9<br>10    | 21.7<br>—    |                 | 21.9<br>—    |                 | dB<br>dB |
| Operating distance                                                                                                           | 2              |              | 10           |              | 30           | 40 <sup>a</sup> | 30           | 40 <sup>a</sup> | km       |
| Channel insertion loss (max)                                                                                                 | 4 <sup>b</sup> |              | 6.3          |              | 15           | 18              | 15           | 18              | dB       |
| Channel insertion loss (min)                                                                                                 | 0              |              |              |              | 10           |                 | 10           |                 | dB       |
| Maximum discrete reflectance                                                                                                 | See 122.11.2.2 |              |              |              |              |                 |              |                 | dB       |
| Allocation for penalties <sup>c</sup> (for maximum TDECQ):<br>for extinction ratio ≥ 4.5 dB<br>for extinction ratio < 4.5 dB | 3.4<br>3.5     | 3.2<br>3.3   | 3.7<br>3.8   | 3.6<br>3.7   | 3.7<br>—     |                 | 3.9<br>—     |                 | dB<br>dB |
| Additional insertion loss allowed                                                                                            | 0              |              | 0            |              | 3            | 0               | 3            | 0               | dB       |

<sup>a</sup> Links longer than 30 km are considered engineered links. Attenuation for such links needs to be less than the worst case for cables containing IEC 60793-2-50 type B1.1, type B1.3, or type B6 a single-mode cabled optical fiber.

<sup>b</sup> The channel insertion loss is calculated using the maximum distance specified in Table 122–8 for 200GBASE-FR4 and 400GBASE-FR8 and fiber attenuation of 0.5 dB/km plus an allocation for connection and splice loss given in 122.11.2.1.

<sup>c</sup> Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

## 122.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

### 122.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 122–15 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 122–15 may be used to perform that test. The test patterns used in this clause are shown in Table 122–14.

**Table 122–14—Test patterns**

| Pattern     | Pattern description             | Defined in   |
|-------------|---------------------------------|--------------|
| Square wave | Square wave (8 threes, 8 zeros) | 120.5.11.2.4 |
| 3           | PRBS31Q                         | 120.5.11.2.2 |
| 4           | PRBS13Q                         | 120.5.11.2.1 |
| 5           | Scrambled idle                  | 119.2.4.9    |
| 6           | SSPRQ                           | 120.5.11.2.3 |

**Table 122–15—Test-pattern definitions and related subclauses**

| Parameter                                                                   | Pattern                                                          | Related subclause |
|-----------------------------------------------------------------------------|------------------------------------------------------------------|-------------------|
| Wavelength                                                                  | Square wave, 3, 4, 5, 6 or valid 200GBASE-R or 400GBASE-R signal | 122.8.2           |
| Side mode suppression ratio                                                 | 3, 5, 6 or valid 200GBASE-R or 400GBASE-R signal                 | 122.8.2           |
| Average optical power                                                       | 3, 5, 6 or valid 200GBASE-R or 400GBASE-R signal                 | 122.8.3           |
| Outer Optical Modulation Amplitude (OMA <sub>outer</sub> )                  | 4 or 6                                                           | 122.8.4           |
| Transmitter and dispersion eye closure for PAM4 (TDECQ)                     | 6                                                                | 122.8.5           |
| Extinction ratio                                                            | 4 or 6                                                           | 122.8.6           |
| Transmitter transition time                                                 | Square wave or 6                                                 | 122.8.7           |
| RIN <sub>17.1</sub> OMA, RIN <sub>15.6</sub> OMA, and RIN <sub>15</sub> OMA | Square wave                                                      | 122.8.8           |
| Stressed receiver conformance test signal calibration                       | 6                                                                | 122.8.10.2        |
| Stressed receiver sensitivity                                               | 3 or 5                                                           | 122.8.10          |

### 122.8.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 122–5 for 200GBASE-FR4, in Table 122–6 for 200GBASE-LR4 and 200GBASE-ER4 and in Table 122–7 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8, if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 122–15.

### 122.8.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 if measured using the methods given in IEC 61280-1-1, with the sum of the optical power from all of the lanes not under test below –30 dBm, per the test setup in Figure 53–6.

#### 122.8.4 Outer Optical Modulation Amplitude ( $OMA_{outer}$ )

The  $OMA_{outer}$  of each lane shall be within the limits given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8. The  $OMA_{outer}$  is measured using a test pattern specified for  $OMA_{outer}$  in Table 122–15. It is the difference between the average optical launch power level  $P_3$ , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level  $P_0$ , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 122–3. For this measurement the sum of the optical power from all of the lanes not under test is below –30 dBm, or if other lanes are operating, a suitable optical filter may be used to separate the lane under test.

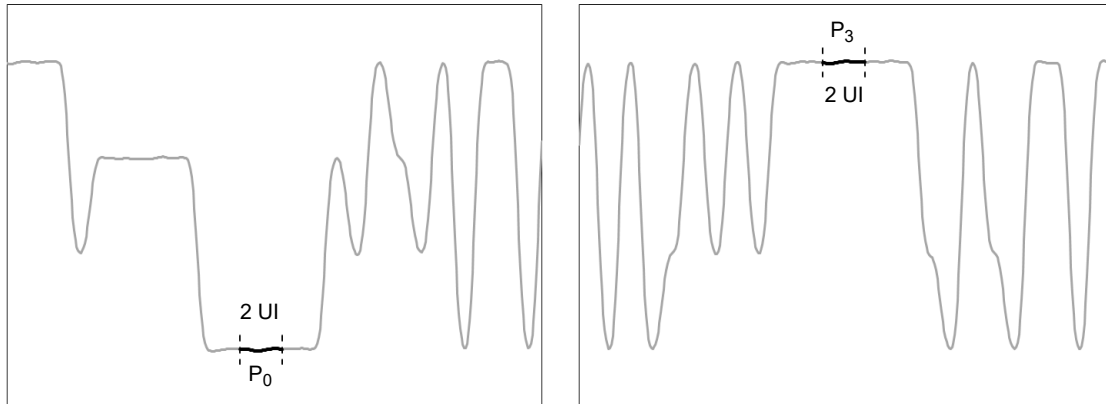


Figure 122–3—Power levels  $P_0$  and  $P_3$  from PRBS13Q test pattern

#### 122.8.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ and  $TDECQ - 10\log_{10}(C_{eq})$  of each lane shall be within the limits given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 if measured using the methods specified in 122.8.5.1, 122.8.5.2, and 122.8.5.3.

TDECQ is a measure of each optical transmitter's vertical eye closure when transmitted through a worst case optical channel (specified in 122.8.5.2), as measured through an optical to electrical converter (O/E) with a bandwidth equivalent to a reference receiver, and equalized with the reference equalizer (as described in 122.8.5.4). The reference receiver and equalizer may be implemented in software or may be part of the oscilloscope.

Table 122–15 specifies the test patterns to be used for measurement of TDECQ.

##### 122.8.5.1 TDECQ conformance test setup

A block diagram for the TDECQ conformance test is shown in Figure 122–4. Other equivalent measurement implementations may be used with suitable calibration.

Each optical lane is tested individually with all other lanes in operation and all lanes using the same test pattern. There shall be at least 31 UI delay between the test pattern on one lane and the pattern on any other lane, so that the symbols on each lane are not correlated within the PMD. The optical splitter and variable reflector are adjusted so that each transmitter is tested with the optical return loss specified in Table 122–16. The state of polarization of the back reflection is adjusted to create the greatest RIN. The optical filter is used to separate the lane under test from the others. Each optical lane is tested with the optical channel described in 122.8.5.2. The combination of the O/E converter and the oscilloscope has a 3 dB bandwidth of

approximately 13.28125 GHz with a fourth-order Bessel-Thomson response to at least  $1.5 \times 26.5625$  GHz, and at frequencies above  $1.5 \times 26.5625$  GHz, the response should not exceed  $-24$  dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.

The optical filter passband ripple shall be limited to 0.5 dB peak-to-peak and the isolation is chosen such that the ratio of the power in the lane being measured to the sum of the powers of all of the other lanes is greater than 20 dB (see ITU-T G.959.1 Annex B).

The test pattern (specified in Table 122–15) is transmitted repetitively by the optical lane under test and the oscilloscope is set up to capture the complete pattern for TDECQ analysis as described in 122.8.5.3. The clock recovery unit (CRU) has a corner frequency of 4 MHz and a slope of 20 dB/decade. The CRU can be implemented in hardware or software depending on oscilloscope technology.

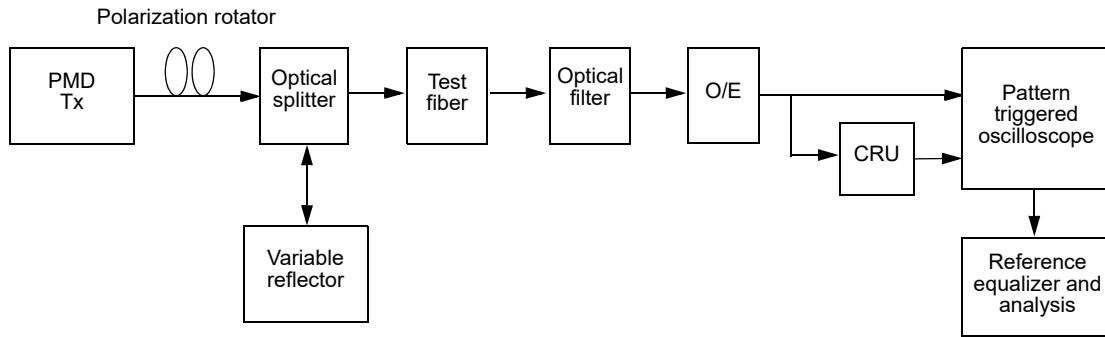


Figure 122–4—TDECQ conformance test block diagram

#### 122.8.5.2 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 122–16.

Table 122–16—Transmitter compliance channel specifications

| PMD type                           | Dispersion <sup>a</sup> (ps/nm)                       |                                                       | Insertion loss <sup>b</sup> | Optical return loss <sup>c</sup> | Max mean DGD |
|------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-----------------------------|----------------------------------|--------------|
|                                    | Minimum                                               | Maximum                                               |                             |                                  |              |
| 200GBASE-FR4<br>or<br>400GBASE-FR8 | $0.0465 \cdot \lambda \cdot [1 - (1324 / \lambda)^4]$ | $0.0465 \cdot \lambda \cdot [1 - (1300 / \lambda)^4]$ | Minimum                     | 17.1 dB                          | 0.8 ps       |
| 200GBASE-LR4<br>or<br>400GBASE-LR8 | $0.2325 \cdot \lambda \cdot [1 - (1324 / \lambda)^4]$ | $0.2325 \cdot \lambda \cdot [1 - (1300 / \lambda)^4]$ | Minimum                     | 15.6 dB                          | 0.8 ps       |
| 200GBASE-ER4<br>or<br>400GBASE-ER8 | $0.93 \cdot \lambda \cdot [1 - (1324 / \lambda)^4]$   | $0.93 \cdot \lambda \cdot [1 - (1300 / \lambda)^4]$   | Minimum                     | 15 dB                            | 0.8 ps       |

<sup>a</sup> The dispersion is measured for the wavelength of the device under test ( $\lambda$  in nm). The coefficient assumes 2 km for 200GBASE-FR4 and 400GBASE-FR8, 10 km for 200GBASE-LR4 and 400GBASE-LR8, and 40 km for 200GBASE-ER4 and 400GBASE-ER8.

<sup>b</sup> There is no intent to stress the sensitivity of the O/E converter associated with the oscilloscope.

<sup>c</sup> The optical return loss is applied at TP2.

A 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 122–16 for the wavelength of the device under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides an optical return loss specified in Table 122–16. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 122–16.

### 122.8.5.3 TDECQ measurement method

TDECQ for 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 is measured as described in 121.8.5.3 with the exception that the reference equalizer is as specified in 122.8.5.4.

### 122.8.5.4 TDECQ reference equalizer

The reference equalizer for 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 is a 5-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period. A functional model of the reference equalizer is shown in Figure 122–5. The sum of the equalizer tap coefficients is equal to 1. Tap 1, tap 2, or tap 3 has the largest magnitude tap coefficient, which is constrained to be at least 0.8.

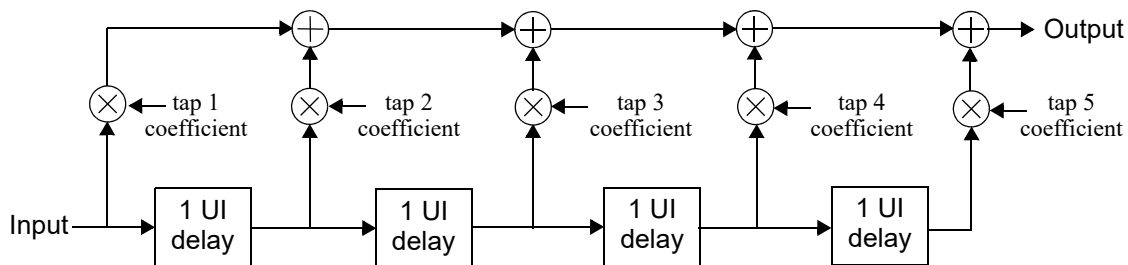


Figure 122–5—TDECQ reference equalizer functional model

NOTE—This reference equalizer is part of the TDECQ test and does not imply any particular receiver equalizer implementation.

### 122.8.6 Extinction ratio

The extinction ratio of each lane shall be within the limits given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 if measured using a test pattern specified for extinction ratio in Table 122–15 with the sum of the optical power from all of the lanes not under test being below –30 dBm, or if other lanes are operating, a suitable optical filter may be used to separate the lane under test. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power level  $P_3$ , measured over the central 2 UI of

a run of 7 threes, and the average optical launch power level  $P_0$ , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 122–3.

### 122.8.7 Transmitter transition time

The transmitter transition time of each lane shall be within the limits given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8, if measured using a test pattern specified for transmitter transition time in Table 122–15.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of  $OMA_{outer}$  to 80% of  $OMA_{outer}$ , or from 80% of  $OMA_{outer}$  to 20% of  $OMA_{outer}$ , for the rising and falling edges respectively, as measured through an O/E converter and oscilloscope with response defined as follows. The combined response of the O/E converter and oscilloscope has a 3 dB bandwidth of approximately 13.28125 GHz with a fourth-order Bessel-Thomson response to at least  $1.5 \times 26.5625$  GHz. At frequencies above  $1.5 \times 26.5625$  GHz, the response should not exceed –24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.

The 0% level and the 100% level are  $P_0$  and  $P_3$  as defined by the  $OMA_{outer}$  measurement procedure (see 121.8.4), with the exception that the square wave test pattern can be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

### 122.8.8 Relative intensity noise ( $RIN_{17.1OMA}$ , $RIN_{15.6OMA}$ , and $RIN_{15OMA}$ )

RIN shall be as defined by the measurement methodology of 52.9.6 with the following exceptions:

- The optical return loss is 17.1 dB for 200GBASE-FR4 and 400GBASE-FR8, 15.6 dB for 200GBASE-LR4 and 400GBASE-LR8, and 15 dB for 200GBASE-ER4 and 400GBASE-ER8.
- Each lane may be tested individually with the sum of the optical power from all of the lanes not under test being below –30 dBm, or if other lanes are operating, a suitable optical filter may be used to separate the lane under test.
- The upper –3 dB limit of the measurement apparatus is to be approximately equal to the signaling rate (i.e., 26.6 GHz).
- The test pattern is according to Table 122–15.

### 122.8.9 Receiver sensitivity

For 200GBASE-FR4, receiver sensitivity is optional and is defined for a transmitter with a value of  $SECQ$  up to 3.1 dB. Receiver sensitivity should meet Equation (122–1), which is illustrated in Figure 122–6.

For 200GBASE-LR4, receiver sensitivity is optional and is defined for a transmitter with a value of  $SECQ$  up to 3.2 dB. Receiver sensitivity should meet Equation (122–2), which is illustrated in Figure 122–6.

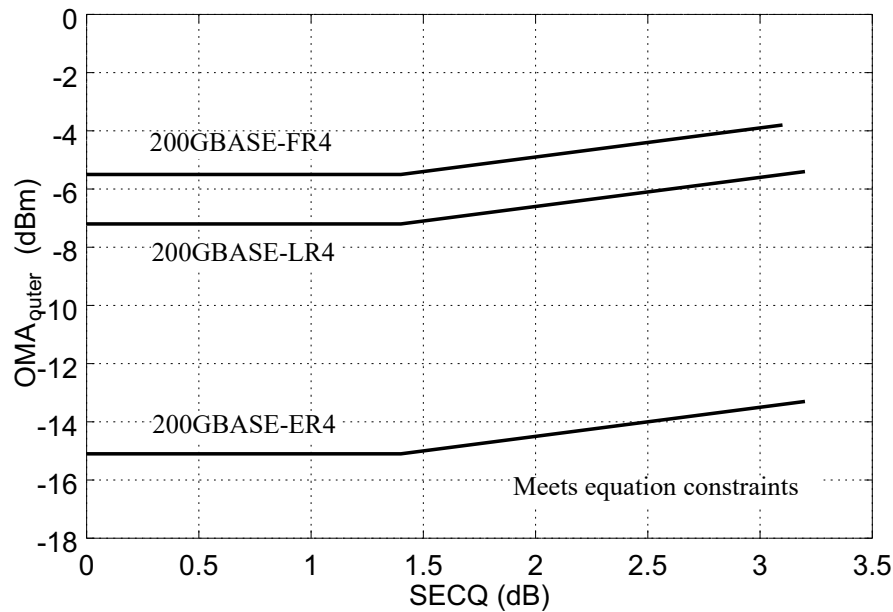
For 200GBASE-ER4, receiver sensitivity is optional and is defined for a transmitter with a value of  $SECQ$  up to 3.2 dB. Receiver sensitivity should meet Equation (122–3), which is illustrated in Figure 122–6.

$$RS = \max(-5.5, SECQ - 6.9) \quad (\text{dBm}) \quad (122-1)$$

$$RS = \max(-7.2, SECQ - 8.6) \quad (\text{dBm}) \quad (122-2)$$

$$RS = \max(-15.1, SECQ - 16.5) \quad (\text{dBm}) \quad (122-3)$$

where  
 $RS$  is the receiver sensitivity  
 $SECQ$  is the SECQ of the transmitter used to measure the receiver sensitivity



**Figure 122-6—Illustration of receiver sensitivity for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4**

For 400GBASE-FR8, receiver sensitivity is optional and is defined for a transmitter with a value of SECQ up to 2.9 dB. Receiver sensitivity should meet Equation (122-4), which is illustrated in Figure 122-7.

For 400GBASE-LR8, receiver sensitivity is optional and is defined for a transmitter with a value of SECQ up to 3.1 dB. Receiver sensitivity should meet Equation (122-5), which is illustrated in Figure 122-7.

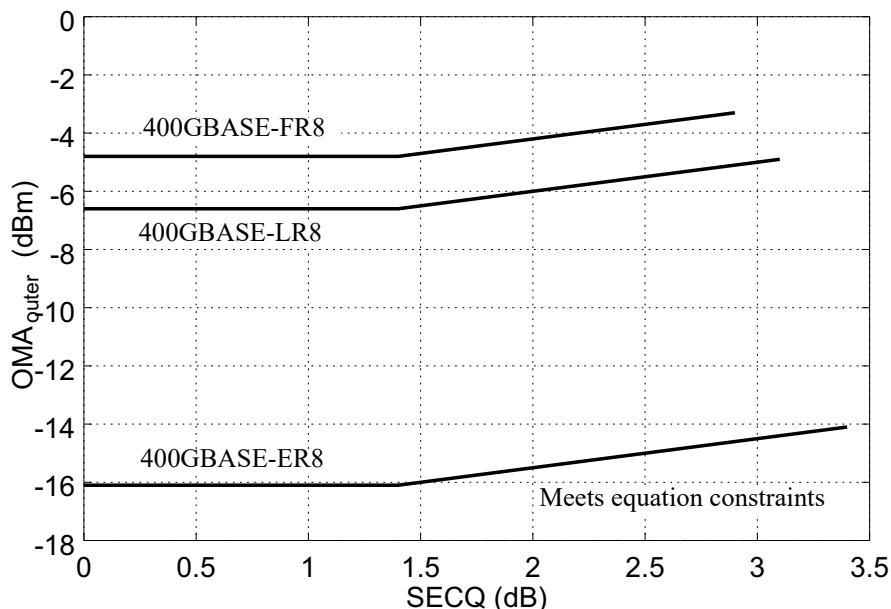
For 400GBASE-ER8, receiver sensitivity is optional and is defined for a transmitter with a value of SECQ up to 3.4 dB. Receiver sensitivity should meet Equation (122-6), which is illustrated in Figure 122-7.

$$RS = \max(-4.8, SECQ - 6.2) \quad (\text{dBm}) \tag{122-4}$$

$$RS = \max(-6.6, SECQ - 8) \quad (\text{dBm}) \tag{122-5}$$

$$RS = \max(-16.1, SECQ - 17.5) \quad (\text{dBm}) \tag{122-6}$$

where  
 $RS$  is the receiver sensitivity  
 $SECQ$  is the SECQ of the transmitter used to measure the receiver sensitivity



**Figure 122-7—Illustration of receiver sensitivity for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8**

The normative requirement for receivers is stressed receiver sensitivity.

#### 122.8.10 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 122-11 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122-12 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 if measured using the method defined in 122.8.10.1 and 122.8.10.3, with the conformance test signal at TP3 as described in 122.8.10.2, using the test pattern specified for SRS in Table 122-15. The BER is required to be met for the lane under test on its own.

Stressed receiver sensitivity is defined with all transmit and receive lanes in operation. Any of the patterns specified for SRS in Table 122-15 is sent from the transmit section of the PMD under test. The signal being transmitted is asynchronous to the received signal.

##### 122.8.10.1 Stressed receiver conformance test block diagram

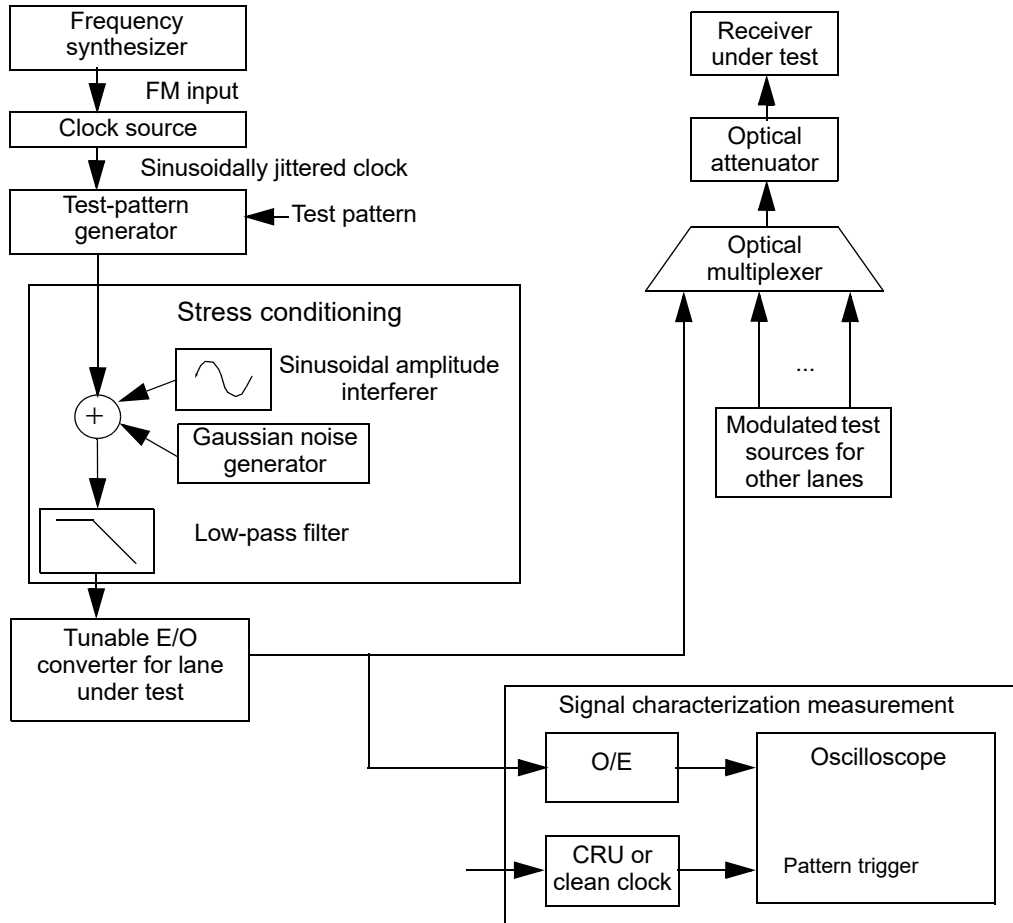
A block diagram for the receiver conformance test is shown in Figure 122-8. The patterns used for the received conformance signal are specified in Table 122-15. The optical test signal is conditioned (stressed) using the stressed receiver methodology defined in 122.8.10.2 and has sinusoidal jitter applied as specified in 121.8.10.4. A suitable test set is needed to characterize the signal used to test the receiver. Stressed receiver conformance test signal verification is described in 122.8.10.3.

The low-pass filter is used to create ISI. The sinusoidal amplitude interferer causes additional eye closure, but in conjunction with the finite edge rates, also causes some jitter.

The sinusoidally jittered clock represents other forms of jitter and also verifies that the receiver under test can track low-frequency jitter. The sinusoidal amplitude interferer may be set at any frequency between

100 MHz and 2 GHz, although care should be taken to avoid harmonic relationships between the sinusoidal interferer, the sinusoidal jitter, the signaling rate, and the pattern repetition rate. The Gaussian noise generator, the amplitude of the sinusoidal interferer, and the low-pass filter are adjusted so that the SECQ specified in Table 122–11 or Table 122–12 is met, according to the methods specified in 122.8.10.2.

For improved visibility for calibration, all elements in the signal path (cables, DC blocks, E/O converter, etc.) should have wide and smooth frequency response, and linear phase response, throughout the spectrum of interest. Baseline wander and overshoot and undershoot should be negligible.



**Figure 122–8—Stressed receiver conformance test block diagram**

### 122.8.10.2 Stressed receiver conformance test signal characteristics and calibration

The stressed receiver conformance test signal characteristics and calibration methods are as described in 121.8.10.2 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 122.8.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.
- An example stressed receiver conformance test setup is shown in Figure 122–8; however, alternative test setups that generate equivalent stress conditions may be used.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the  $RIN_{xx,x}$ OMA of the SRS test source should be no greater than the value specified in Table 122–9 for

200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.

- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 122–9 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–10 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.
- The required values of the “Stressed receiver sensitivity ( $\text{OMA}_{\text{outer}}$ ), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test”, “SECQ –  $10\log_{10}(C_{\text{eq}})$  (max), lane under test”, and “ $\text{OMA}_{\text{outer}}$  of each aggressor lane” are as given in Table 122–11 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4 and in Table 122–12 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.

### 122.8.10.3 Stressed receiver conformance test signal verification

The SECQ of the stressed receiver conformance test signal is measured according to 122.8.5, except that the test fiber is not used. The clock output from the clock source in Figure 122–8 is modulated with the sinusoidal jitter. To use an oscilloscope to calibrate the final stressed eye jitter that includes the sinusoidal jitter component, a separate clock source (clean clock of Figure 122–8) is required that is synchronized to the source clock, but not modulated with the jitter source.

Care should be taken when characterizing the test signal because excessive noise/jitter in the measurement system would result in an input signal that does not fully stress the receiver under test. Running the receiver tolerance test with a signal that is under-stressed may result in the deployment of non-compliant receivers. The noise/jitter introduced by the O/E, filters, and oscilloscope should be negligible or the results should be corrected for its effects. While the details of test equipment are beyond the scope of this standard, it is recommended that the implementer fully characterize the test equipment and apply appropriate guard bands so that the stressed receiver conformance input signal meets the stress and sinusoidal jitter specified in 122.8.10.2 and 121.8.10.4.

## 122.9 Safety, installation, environment, and labeling

### 122.9.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

### 122.9.2 Laser safety

200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, and 400GBASE-LR8 optical transceivers shall conform to Hazard Level 1 laser requirements and 400GBASE-ER8 optical transceivers shall conform to Hazard Level 1M laser requirements, as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.<sup>210</sup>

<sup>210</sup>A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

### 122.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

### 122.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

### 122.9.5 Electromagnetic emission

A system integrating a 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

### 122.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

### 122.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 400GBASE-FR8).

Labeling requirements for Hazard Level 1 and Hazard Level 1M lasers are given in the laser safety standards referenced in 122.9.2.

## 122.10 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 122–9.

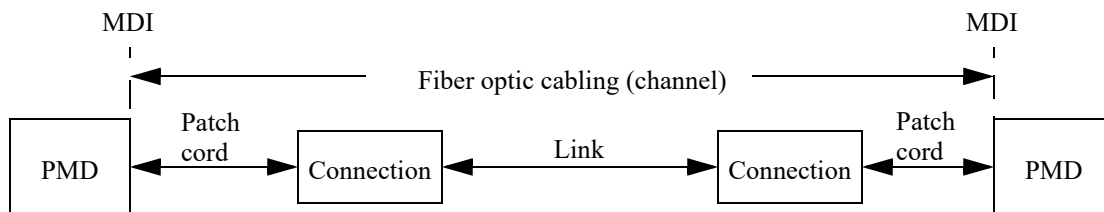


Figure 122–9—Fiber optic cabling model

The channel insertion loss is given in Table 122–17. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term channel is used here for consistency with generic cabling standards.

**Table 122–17—Fiber optic cabling (channel) characteristics**

| Description                                 | 200GBASE-FR4 | 400GBASE-FR8 | 200GBASE-LR4     | 400GBASE-LR8 | 200GBASE-ER4    |      | 400GBASE-ER8 |      | Unit  |
|---------------------------------------------|--------------|--------------|------------------|--------------|-----------------|------|--------------|------|-------|
| Operating distance (max)                    | 2            |              | 10               |              | 30              | 40   | 30           | 40   | km    |
| Channel insertion loss <sup>a,b</sup> (max) | 4            |              | 6.3 <sup>c</sup> |              | 18              |      | 18           |      | dB    |
| Channel insertion loss (min)                | 0            |              | 0                |              | 10 <sup>d</sup> |      |              |      | dB    |
| Positive dispersion <sup>b</sup> (max)      | 6.7          | 1.9          | 9.5              |              | 28              | 37   | 28           | 37   | ps/nm |
| Negative dispersion <sup>b</sup> (min)      | −11.9        | −10.2        | −28.4            | −50.8        | −85             | −114 | −151         | −201 | ps/nm |
| DGD_max <sup>e</sup>                        | 3            |              | 8                |              | 10.3            |      |              |      | ps    |
| Optical return loss (min)                   | 25           |              | 22               |              | 19              |      |              |      | dB    |

<sup>a</sup> These channel insertion loss values include cable, connectors, and splices.

<sup>b</sup> Over the wavelength range 1264.5 nm to 1337.5 nm for 200GBASE-FR4, 1294.53 nm to 1310.19 nm for 200GBASE-LR4 and 200GBASE-ER4, and 1272.55 nm to 1310.19 nm for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8.

<sup>c</sup> Using 0.46 dB/km at 1272.55 nm attenuation for optical fiber cables derived from Appendix I of ITU-T G.695 may not support operation at 10 km for 400GBASE-LR8 under worst case conditions.

<sup>d</sup> Channel insertion loss (min) may be implemented with an optical attenuator.

<sup>e</sup> Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD\_max is the maximum differential group delay that the system is required to tolerate.

## 122.11 Characteristics of the fiber optic cabling (channel)

The 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 fiber optic cabling shall meet the specifications defined in Table 122–17. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

### 122.11.1 Optical fiber cable

The fiber optic cable requirements are satisfied by cables containing IEC 60793-2-50 type B1.1 (dispersion un-shifted single-mode), type B1.3 (low water peak single-mode), or type B6\_a (bend insensitive) fibers or the requirements in Table 122–18 where they differ.

### 122.11.2 Optical fiber connection

An optical fiber connection, as shown in Figure 122–9, consists of a mated pair of optical connectors.

**Table 122–18—Optical fiber and cable characteristics**

| Description                                | Value                                 | Unit                  |
|--------------------------------------------|---------------------------------------|-----------------------|
| Nominal fiber specification wavelength     | 1310                                  | nm                    |
| Cabled optical fiber attenuation (max)     | 0.47 <sup>a</sup> or 0.5 <sup>b</sup> | dB/km                 |
| Zero dispersion wavelength ( $\lambda_0$ ) | $1300 \leq \lambda_0 \leq 1324$       | nm                    |
| Dispersion slope (max) ( $S_0$ )           | 0.093                                 | ps/nm <sup>2</sup> km |

<sup>a</sup> The 0.47 dB/km at 1264.5 nm attenuation for optical fiber cables is derived from Appendix I of ITU-T G.695.

<sup>b</sup> The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3. Using 0.5 dB/km may not support operation at 10 km for 200GBASE-LR4 or 400GBASE-LR8 or at 40 km for 200GBASE-ER4 or 400GBASE-ER8.

### 122.11.2.1 Connection insertion loss

The maximum link distance for 200GBASE-LR4, 200GBASE-ER4, 400GBASE-LR8, and 400GBASE-ER8 is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB. The maximum link distance for 200GBASE-FR4 and 400GBASE-FR8 is based on an allocation of 3 dB total connection and splice loss. Connections with different loss characteristics may be used provided the requirements of Table 122–17 are met.

### 122.11.2.2 Maximum discrete reflectance

The maximum value for each discrete reflectance shall be less than or equal to the value shown in Table 122–19 corresponding to the number of discrete reflectances above –55 dB within the channel. For numbers of discrete reflectances in between two numbers shown in the table, the lower of the two corresponding maximum discrete reflectance values applies.

**Table 122–19—Maximum value of each discrete reflectance**

| Number of discrete reflectances above –55 dB | Maximum value for each discrete reflectance |                              |                              |
|----------------------------------------------|---------------------------------------------|------------------------------|------------------------------|
|                                              | 200GBASE-FR4 or 400GBASE-FR8                | 200GBASE-LR4 or 400GBASE-LR8 | 200GBASE-ER4 or 400GBASE-ER8 |
| 1                                            | –25 dB                                      | –22 dB                       | –19 dB                       |
| 2                                            | –31 dB                                      | –29 dB                       | –27 dB                       |
| 4                                            | –35 dB                                      | –33 dB                       | –32 dB                       |
| 6                                            | –38 dB                                      | –35 dB                       | –35 dB                       |
| 8                                            | –40 dB                                      | –37 dB                       | –37 dB                       |
| 10                                           | –41 dB                                      | –39 dB                       | –39 dB                       |

### 122.11.3 Medium Dependent Interface (MDI) requirements

The 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, or 400GBASE-ER8 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 122–9). Examples of an MDI include the following:

- a) Connectorized fiber pigtail
- b) PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1-1 and IEC 61753-021-2.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 122.5.1, not at the MDI.

### 122.12 Requirements for interoperation between 200GBASE-ER4 and 200GBASE-LR4

The 200GBASE-ER4 and 200GBASE-LR4 PMDs can interoperate with each other (over an engineered link) provided that the fiber optic cabling (channel) characteristics for 200GBASE-LR4 are met, with the exception of the maximum and minimum channel insertion loss values, which are given in Table 122–20 for the two link directions separately. Attenuators may be used to achieve the required losses.

**Table 122–20—Channel insertion loss requirements for interoperation between 200GBASE-ER4 and 200GBASE-LR4**

| Direction                                         | Min loss | Max loss | Unit |
|---------------------------------------------------|----------|----------|------|
| 200GBASE-ER4 transmitter to 200GBASE-LR4 receiver | 2.3      | 10.1     | dB   |
| 200GBASE-LR4 transmitter to 200GBASE-ER4 receiver | 8.7      | 14.2     | dB   |

### 122.13 Requirements for interoperation between 400GBASE-ER8 and 400GBASE-FR8

The 400GBASE-ER8 and 400GBASE-FR8 PMDs can interoperate with each other (over an engineered link) provided that the fiber optic cabling (channel) characteristics for 400GBASE-FR8 are met, with the exception of the maximum and minimum channel insertion loss values, which are given in Table 122–21 for the two link directions separately. Attenuators may be used to achieve the required losses.

**Table 122–21—Channel insertion loss requirements for interoperation between 400GBASE-ER8 and 400GBASE-FR8**

| Direction                                         | Min loss | Max loss | Unit |
|---------------------------------------------------|----------|----------|------|
| 400GBASE-ER8 transmitter to 400GBASE-FR8 receiver | 0.7      | 6.9      | dB   |
| 400GBASE-FR8 transmitter to 400GBASE-ER8 receiver | 9.7      | 15.1     | dB   |

## 122.14 Requirements for interoperation between 400GBASE-ER8 and 400GBASE-LR8

The 400GBASE-ER8 and 400GBASE-LR8 PMDs can interoperate with each other (over an engineered link) provided that the fiber optic cabling (channel) characteristics for 400GBASE-LR8 are met, with the exception of the maximum and minimum channel insertion loss values, which are given in Table 122–22 for the two link directions separately. Attenuators may be used to achieve the required losses.

**Table 122–22—Channel insertion loss requirements for interoperation between 400GBASE-ER8 and 400GBASE-LR8**

| Direction                                         | Min loss | Max loss | Unit |
|---------------------------------------------------|----------|----------|------|
| 400GBASE-ER8 transmitter to 400GBASE-LR8 receiver | 0.7      | 8.5      | dB   |
| 400GBASE-LR8 transmitter to 400GBASE-ER8 receiver | 9.7      | 15.8     | dB   |

## 122.15 Protocol implementation conformance statement (PICS) proforma for Clause 122, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8<sup>211</sup>

### 122.15.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 122, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 122.15.2 Identification

#### 122.15.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                                                     |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                                               |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                                             |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                                                                                                                                                                                |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                                                 |  |
| <p>NOTE 1—Required for all implementations.<br/>         NOTE 2—May be completed as appropriate in meeting the requirements for the identification.<br/>         NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).</p> |  |

#### 122.15.2.2 Protocol summary

|                                                                                                                                                                                                                             |                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                                                                         | IEEE Std 802.3-2022, Clause 122, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                                                                             |                                                                                                                                                                                   |
| <p>Have any Exception items been required? No <input type="checkbox"/> Yes <input type="checkbox"/><br/>         (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)</p> |                                                                                                                                                                                   |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>211</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

### 122.15.3 Major capabilities/options

| Item | Feature                                               | Subclause | Value/Comment                                                                                                      | Status | Support           |
|------|-------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|--------|-------------------|
| *FR4 | 200GBASE-FR4 PMD                                      | 122.7     | Device supports requirements for 200GBASE-FR4 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *LR4 | 200GBASE-LR4 PMD                                      | 122.7     | Device supports requirements for 200GBASE-LR4 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *ER4 | 200GBASE-ER4 PMD                                      | 122.7     | Device supports requirements for 200GBASE-ER4 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *FR8 | 400GBASE-FR8 PMD                                      | 122.7     | Device supports requirements for 400GBASE-FR8 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *LR8 | 400GBASE-LR8 PMD                                      | 122.7     | Device supports requirements for 400GBASE-LR8 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *ER8 | 400GBASE-ER8 PMD                                      | 122.7     | Device supports requirements for 400GBASE-ER8 PHY                                                                  | O.1    | Yes [ ]<br>No [ ] |
| *INS | Installation / cable                                  | 122.10    | Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer | O      | Yes [ ]<br>No [ ] |
| TP1  | Reference point TP1 exposed and available for testing | 122.5.1   | This point may be made available for use by implementers to certify component conformance                          | O      | Yes [ ]<br>No [ ] |
| TP4  | Reference point TP4 exposed and available for testing | 122.5.1   | This point may be made available for use by implementers to certify component conformance                          | O      | Yes [ ]<br>No [ ] |
| DC   | Delay constraints                                     | 122.5.1   | Device conforms to delay constraints                                                                               | M      | Yes [ ]           |
| SC   | Skew constraints                                      | 122.3.2   | Device conforms to Skew and Skew Variation constraints                                                             | M      | Yes [ ]           |
| *MD  | MDIO capability                                       | 122.4     | Registers and interface supported                                                                                  | O      | Yes [ ]<br>No [ ] |

## 122.15.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8

### 122.15.4.1 PMD functional specifications

| Item | Feature                                                                 | Subclause | Value/Comment                                                                                      | Status | Support                      |
|------|-------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|--------|------------------------------|
| F1   | Compatible with 200GBASE-R or 400GBASE-R PCS and PMA                    | 122.1     |                                                                                                    | M      | Yes [ ]                      |
| F2   | Integration with management functions                                   | 122.1     |                                                                                                    | O      | Yes [ ]<br>No [ ]            |
| F3   | Bit error ratio                                                         | 122.1.1   | Meets the BER specified in 122.1.1                                                                 | M      | Yes [ ]                      |
| F4   | Transmit function                                                       | 122.5.2   | Conveys symbols from PMD service interface to MDI                                                  | M      | Yes [ ]                      |
| F5   | Optical multiplexing and delivery to the MDI                            | 122.5.2   | Optically multiplexes the four or eight optical signals for delivery to the MDI                    | M      | Yes [ ]                      |
| F6   | Mapping between optical signal and logical signal for transmitter       | 122.5.2   | Highest optical power corresponds to tx_symbol = three                                             | M      | Yes [ ]                      |
| F7   | Receive function                                                        | 122.5.3   | Conveys symbols from MDI to PMD service interface                                                  | M      | Yes [ ]                      |
| F8   | Conversion of four or eight optical signals to eight electrical signals | 122.5.3   | For delivery to the PMD service interface                                                          | M      | Yes [ ]                      |
| F9   | Mapping between optical signal and logical signal for receiver          | 122.5.3   | Highest optical power corresponds to rx_symbol = three                                             | M      | Yes [ ]                      |
| F10  | Global Signal Detect function                                           | 122.5.4   | Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_DETECT)           | M      | Yes [ ]                      |
| F11  | Global Signal Detect behavior                                           | 122.5.4   | SIGNAL_DETECT is a global indicator of the presence of optical signals on all four or eight lanes  | M      | Yes [ ]                      |
| F12  | Lane-by-lane Signal Detect function                                     | 122.5.5   | Sets PMD_signal_detect <sub>i</sub> values on a lane-by-lane basis per requirements of Table 122–4 | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| F13  | PMD reset function                                                      | 122.5.6   | Resets the PMD sublayer                                                                            | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 122.15.4.2 Management functions

| Item | Feature                                    | Subclause | Value/Comment                                                                                             | Status | Support                      |
|------|--------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------|--------|------------------------------|
| M1   | Management register set                    | 122.4     |                                                                                                           | MD:M   | Yes [ ]<br>N/A [ ]           |
| M2   | Global transmit disable function           | 122.5.7   | Disables all of the optical transmitters with the PMD_global_transmit_disable variable                    | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| M3   | PMD lane-by-lane transmit disable function | 122.5.8   | Disables the optical transmitter via MDIO on the lane associated with the PMD_transmit_disable_i variable | MD:O.2 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| M4   | PMD lane-by-lane transmit disable          | 122.5.8   | Disables each optical transmitter independently if M3 = No                                                | O.2    | Yes [ ]<br>N/A [ ]           |
| M5   | PMD_fault function                         | 122.5.9   | Sets PMD_fault to one if any local fault is detected                                                      | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| M6   | PMD_transmit_fault function                | 122.5.10  | Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane                          | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| M7   | PMD_receive_fault function                 | 122.5.11  | Sets PMD_receive_fault to one if a local fault is detected on any receive lane                            | MD:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 122.15.4.3 PMD to MDI optical specifications for 200GBASE-FR4

| Item | Feature                                         | Subclause | Value/Comment            | Status | Support            |
|------|-------------------------------------------------|-----------|--------------------------|--------|--------------------|
| FRF1 | Transmitter meets specifications in Table 122–9 | 122.7.1   | Per definitions in 122.8 | FR4:M  | Yes [ ]<br>N/A [ ] |
| FRF2 | Receiver meets specifications in Table 122–11   | 122.7.2   | Per definitions in 122.8 | FR4:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.4 PMD to MDI optical specifications for 200GBASE-LR4

| Item | Feature                                         | Subclause | Value/Comment            | Status | Support            |
|------|-------------------------------------------------|-----------|--------------------------|--------|--------------------|
| LRF1 | Transmitter meets specifications in Table 122–9 | 122.7.1   | Per definitions in 122.8 | LR4:M  | Yes [ ]<br>N/A [ ] |
| LRF2 | Receiver meets specifications in Table 122–11   | 122.7.2   | Per definitions in 122.8 | LR4:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.5 PMD to MDI optical specifications for 200GBASE-ER4

| Item | Feature                                         | Subclause | Value/Comment            | Status | Support            |
|------|-------------------------------------------------|-----------|--------------------------|--------|--------------------|
| ERF1 | Transmitter meets specifications in Table 122–9 | 122.7.1   | Per definitions in 122.8 | ER4:M  | Yes [ ]<br>N/A [ ] |
| ERF2 | Receiver meets specifications in Table 122–11   | 122.7.2   | Per definitions in 122.8 | ER4:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.6 PMD to MDI optical specifications for 400GBASE-FR8

| Item | Feature                                          | Subclause | Value/Comment            | Status | Support            |
|------|--------------------------------------------------|-----------|--------------------------|--------|--------------------|
| FRE1 | Transmitter meets specifications in Table 122–10 | 122.7.1   | Per definitions in 122.8 | FR8:M  | Yes [ ]<br>N/A [ ] |
| FRE2 | Receiver meets specifications in Table 122–12    | 122.7.2   | Per definitions in 122.8 | FR8:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.7 PMD to MDI optical specifications for 400GBASE-LR8

| Item | Feature                                          | Subclause | Value/Comment            | Status | Support            |
|------|--------------------------------------------------|-----------|--------------------------|--------|--------------------|
| LRE1 | Transmitter meets specifications in Table 122–10 | 122.7.1   | Per definitions in 122.8 | LR8:M  | Yes [ ]<br>N/A [ ] |
| LRE2 | Receiver meets specifications in Table 122–12    | 122.7.2   | Per definitions in 122.8 | LR8:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.8 PMD to MDI optical specifications for 400GBASE-ER8

| Item | Feature                                          | Subclause | Value/Comment            | Status | Support            |
|------|--------------------------------------------------|-----------|--------------------------|--------|--------------------|
| ERE1 | Transmitter meets specifications in Table 122–10 | 122.7.1   | Per definitions in 122.8 | ER8:M  | Yes [ ]<br>N/A [ ] |
| ERE2 | Receiver meets specifications in Table 122–12    | 122.7.2   | Per definitions in 122.8 | ER8:M  | Yes [ ]<br>N/A [ ] |

#### 122.15.4.9 Optical measurement methods

| Item | Feature                                                 | Subclause | Value/Comment                                | Status | Support |
|------|---------------------------------------------------------|-----------|----------------------------------------------|--------|---------|
| OM1  | Measurement cable                                       | 122.8     | 2 m to 5 m in length                         | M      | Yes [ ] |
| OM2  | Center wavelength and SMSR                              | 122.8.2   | Per IEC 61280-1-3 under modulated conditions | M      | Yes [ ] |
| OM3  | Average optical power                                   | 122.8.3   | Per IEC 61280-1-1                            | M      | Yes [ ] |
| OM4  | OMA measurements                                        | 122.8.4   | Each lane                                    | M      | Yes [ ] |
| OM5  | Transmitter and dispersion eye closure for PAM4 (TDECQ) | 122.8.5   | Each lane                                    | M      | Yes [ ] |
| OM6  | Extinction ratio                                        | 122.8.6   | Each lane                                    | M      | Yes [ ] |
| OM7  | Transmitter transition time                             | 122.8.7   | Each lane                                    | M      | Yes [ ] |
| OM8  | RIN <sub>xx</sub> OMA measurement procedure             | 122.8.8   | Each lane                                    | M      | Yes [ ] |
| OM9  | Stressed receiver sensitivity                           | 122.8.10  | Each lane                                    | M      | Yes [ ] |

#### 122.15.4.10 Environmental specifications

| Item | Feature                          | Subclause | Value/Comment                                                                                        | Status | Support            |
|------|----------------------------------|-----------|------------------------------------------------------------------------------------------------------|--------|--------------------|
| ES1  | General safety                   | 122.9.1   | Conforms to J.2                                                                                      | M      | Yes [ ]            |
| ES2  | Laser safety—IEC Hazard Level 1  | 122.9.2   | Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2                 | !ER8:M | Yes [ ]<br>N/A [ ] |
| ES3  | Laser safety—IEC Hazard Level 1M | 122.9.2   | Conforms to Hazard Level 1M laser requirements defined in IEC 60825-1 and IEC 60825-2                | ER8:M  | Yes [ ]<br>N/A [ ] |
| ES4  | Electromagnetic interference     | 122.9.5   | Complies with applicable local and national codes for the limitation of electromagnetic interference | M      | Yes [ ]            |

#### 122.15.4.11 Characteristics of the fiber optic cabling and MDI

| Item | Feature                      | Subclause  | Value/Comment                                | Status | Support            |
|------|------------------------------|------------|----------------------------------------------|--------|--------------------|
| OC1  | Fiber optic cabling          | 122.11     | Meets requirements specified in Table 122–17 | INS:M  | Yes [ ]<br>N/A [ ] |
| OC2  | Maximum discrete reflectance | 122.11.2.2 | Meets requirements specified in Table 122–19 | INS:M  | Yes [ ]<br>N/A [ ] |
| OC3  | MDI requirements             | 122.11.3   | Meets IEC 61753-1-1 and IEC 61753-021-2      | INS:M  | Yes [ ]<br>N/A [ ] |